

ADVANCED LINEAR ALGEBRA

HOMEWORK 1

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P.A.1 Show that:

- (a) $(1 + i)^4 = -4$
- (b) $(1 - i)^{-1} - (1 + i)^{-i} = i$
- (c) $(i - 1)^{-4} = -1/4$
- (d) $10(1 + 3i)^{-1} = 1 - 3i$
- (e) $(\sqrt{3} + i)^3 = 8i$

Solution. (a) Consider the following

□

P.A.3 Let $z = 3+4i$. Plot and label the following points in the complex plane: $z, -z, \bar{z}, -\bar{z}, 1/z, 1/\bar{z}, -1/\bar{z}$

Solution. something

□

P.A.6 Write the following in polar form:))

- (1) $1 + i$
- (2) $(1 + i)^2$
- (3) $(1 + i)^4$